Fei Xue, Ph.D.

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Contents

I. Executive Summary

II. General Information

A. Education

Ph.D., 2019	Statistics, University of Illinois Urbana-Champaign, Champaign, IL, USA. Advisor: Annie Qu Dissertation: *Variable selection for high-dimensional complex data*
B.S., 2014	Mathematics and Applied Mathematics, Fudan University, Shanghai, China.

B. Previous academic appointments

2019 - 2021 Postdoc Researcher, Department of Biostatistics, Epidemiology and Informatics, University of Pennsylvania, Philadelphia, PA, USA.
Advisor: Hongzhe Li

C. Present academic appointment

2021 - now Assistant Professor, Department of Statistics, Purdue University, West Lafayette, IN.

D. Industrial, business, and government positions

E. Awards and Honors

- External to Purdue
  - 2022 IMS New Researchers Travel Award, Institute of Mathematical Statistics, USA.
  - 2019 ICSA New Researcher Award, International Chinese Statistical Association, USA.
  - 2019 IMS Hannan Graduate Student Travel Award, Institute of Mathematical Statistics, USA.
  - 2018 ASA Student Paper Award, Statistical Learning and Data Science Section of the American Statistical Association, USA.
  - 2018 Horace W. Norton Prize for Outstanding Thesis Research Finalist, Department of Statistics, UIUC, USA.
- Internal to Purdue
  - 2026 Outstanding Assistant Professor Award, Department of Statistics, College of Science, Purdue University, USA.
  - 2024 Seed for Success Acorn Award, Purdue University, USA.

F. Memberships in academic, professional, and scholarly societies

- Member of American Statistical Association (ASA)
- Member of Institute of Mathematical Statistics (IMS)
- Member of International Chinese Statistical Association (ICSA)

G. Other items unique to the person or Department not included in A-D.

III. Discovery

A. Discussion

Dr. Xue's research develops statistical methodology for complex, heterogeneous, and high-dimensional data arising in genomics, mobile health, and biomedical sciences. A central theme is integrating information from multiple data sources under constraints such as missingness, high dimensionality, and computational scalability. Her contributions span four interconnected areas: multi-source data integration and missing data, statistical genetics and genomics, mobile health and wearable device data, and causal mediation analysis. Her work has appeared in leading journals including the *Annals of Statistics*, *Journal of the American Statistical Association* (JASA), *Biometrika*, and *Annals of Applied Statistics*, supported by grants from the National Science Foundation and the National Institutes of Health.

Multi-Source Data Integration and Missing Data. With the proliferation of multisource and multimodality data, block-wise missingness, where entire measurement blocks are absent for a subset of subjects, has become a central analytical challenge across biomedicine and the social sciences. Dr. Xue developed a multiple block-wise imputation (MBI) approach that constructs estimating equations over all possible missing patterns and combines them via the generalized method of moments, achieving superior prediction accuracy on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset compared to existing methods (*JASA*, 2021). Building on the MBI framework, she proposed a multiple-imputed Dantzig selector (MDS) for high-dimensional inference under a semi-supervised setting in which the outcome is observed for only part of the sample. A bias-corrected MDS estimator derived through projected estimating equations enjoys asymptotic normality and enables valid confidence interval construction (*Statistica Sinica*, 2025). Together, these works provide methodologically rigorous foundations for regression and inference with block-wise missing data in high-dimensional settings.

More recently, Dr. Xue developed identifiable deep latent variable models for data with missing-not-at-random (MNAR) mechanisms, under revision at the *Journal of Machine Learning Research*. Under a conditional no self-censoring assumption given latent variables, this work establishes the first nonparametric identifiability guarantee for deep generative models under non-ignorable missingness, which is a fundamental theoretical contribution as prior deep learning methods for MNAR data largely ignored this issue. A new importance-weighted autoencoder algorithm is proposed for MNAR imputation and data generation, demonstrating advantages over state-of-the-art methods on HIV clinical data and music recommendation datasets. In collaborative work, Dr. Xue also developed an empirical Bayes framework for multi-response regression that derives an asymptotically optimal linear shrinkage estimator under relative savings loss without requiring sparsity or low-rank structural assumptions, which are difficult to verify in practice (accepted, *Statistica Sinica*). Applied to tissue-wide association studies, this proposed method achieves superior multi-tissue gene expression prediction performance.

Statistical Genetics and Genomics. Dr. Xue has made significant contributions to multi-tissue genomics using the Genotype-Tissue Expression (GTEx) project data, which contains RNA-seq data from over 45 tissues of more than 800 donors. She developed an empirical Bayes regression model for predicting the effects of single nucleotide polymorphisms (SNPs) on gene expression across multiple tissues simultaneously (*Statistica Sinica*, 2026). The model employs a mixture prior, which combines a multivariate Gaussian distribution with a point mass at zero, to accommodate tissue-specific genetic effects, allowing SNPs to influence expression in only a subset of tissues. The method asymptotically achieves minimum Bayes risk, outperforms competing multi-tissue approaches in gene expression prediction on held-out GTEx data, and reveals that while genetic effects are broadly shared, effect sizes could vary substantially across tissues. This finding has direct implications for transcriptome-wide association studies (TWAS), which rely on accurate SNP-based gene expression prediction.

A complementary theoretical contribution of Dr. Xue establishes rigorous foundations for linkage disequilibrium score regression (LDSC), one of the most widely used tools for genome-wide association study (GWAS) summary statistics analysis (*Annals of Statistics*, 2025). Working within a fixed-effect data integration framework, Dr. Xue accounted for genome-wide dependence among high-dimensional GWAS summary statistics and block-diagonal dependence in estimated LD scores. She clarified the precise conditions under which asymptotic normality of LDSC-based estimators is achievable, which is a practically important result given LDSC's central role in heritability and genetic correlation estimation. She also extended LDSC to cross-ancestry analyses by modeling LD pattern disparities across global populations, with validation using UK Biobank data. In addition, Dr. Xue has contributed to the BIGA cloud computing platform for bivariate cross-trait genetic architecture analysis (accepted, *JASA*), and to PennPRS, a centralized platform for polygenic risk score training that provides a public resource for the precision medicine community.

Mobile Health and Wearable Device Data. Mobile health data pose significant analytical challenges: they are heterogeneous, irregularly sampled, multi-resolution, and longitudinal. Dr. Xue proposed an individualized dynamic latent factor model for integrating multiple irregular multimodality time series (*Biometrika*, 2024). The model maps multi-resolution observations to a shared latent space and captures heterogeneous longitudinal dynamics through individualized latent factors, enabling interpolation of unsampled time series. The paper provides theoretical bounds on integrated interpolation error and B-spline approximation convergence rates, and demonstrates superior performance on smartwatch data. This work has broad applicability to mobile health monitoring studies where physiological signals are sampled at different rates and with irregular timing.

She has extended this research through a generalized heterogeneous functional model for large-scale mobile health data (under revision, *JASA*) that uses a pairwise fusion penalty to identify latent subgroups with distinct time-varying functional effects of physical activity on disease outcomes, with a pre-clustering procedure for computational scalability. Applied to over 79,000 UK Biobank participants, the method uncovers subgroup-specific activity timing effects on mental disorders and Parkinson's disease risk. In a more recent project, Dr. Xue developed a framework for early Parkinson's disease detection from wrist-worn accelerometry, integrating pretrained sleep staging and physical activity recognition models with hierarchical nonstationary Markov chains for behavioral transition modeling, and gradient-boosted Cox regression for time-to-diagnosis prediction. This work demonstrates the potential of combining pretrained deep learning models, temporal sequence modeling, and survival analysis to detect prodromal behavioral signatures of neurodegeneration at population scale.

Causal Mediation Analysis. Dr. Xue has built a coherent program in causal mediation methodology addressing the analytical demands of modern biomedical data. Her heterogeneous mediation analysis framework allows mediation effects to vary across latent subpopulations, providing a more flexible and scientifically meaningful alternative to standard homogeneous approaches (*JASA*, 2022). Applied to a predominantly African American cohort, this method uncovered heterogeneous epigenomic pathways linking traumatic stress, DNA methylation, and post-traumatic stress disorder (PTSD) risk. Identifying subpopulation-specific mediation pathways is of direct importance to precision psychiatry, as it reveals differential biological mechanisms that are obscured in population-averaged analyses.

She extended this work to longitudinal settings through a time-varying mediation analysis framework for incomplete data (*Annals of Applied Statistics*, 2025), combining structural equation modeling with multiple blockwise imputation and generalized method of moments estimation to handle high-dimensional repeated measurements and non-monotone missingness. This advance is particularly valuable for longitudinal epigenomic studies, where mediation effects may evolve over time and missing data are common. In more recent collaborative work, Dr. Xue contributed to joint estimators for treatment effects on multiple correlated omics outcomes with false discovery rate (FDR) control, accounting for outcome dependence in randomized studies. A complementary application investigates causal pathways from depressive symptom profiles to Alzheimer's disease risk using ADNI data, finding that memory-related depressive symptoms affect cognitive decline through hippocampal volume and cerebral glucose metabolism. Together, Dr. Xue's causal inference contributions span mediation methodology, multi-outcome treatment effect estimation, and causal pathway characterization in omics and neuroimaging research.

B. Publications

According to "SCImago Journal Rank (SJR)" (`https://www.scimagojr.com/journalrank.php?category=2613&area=2600`), a list of the top-tier journals in **Statistics and Probability** is as follows:

- Annals of Mathematics
- Annals of Statistics
- Journal of the Royal Statistical Society. Series B: Statistical Methodology
- Journal of the American Statistical Association
- Journal of Business and Economic Statistics
- Biometrika

1. Refereed

1. **Fei Xue**, Hongzhe Li (2026). An empirical Bayes regression for multi-tissue eQTL data analysis. ***Statistica Sinica***, 36, 415-437.

2. **Fei Xue**, Bingxin Zhao (2025). High-dimensional statistical inference for linkage disequilibrium score regression and its cross-ancestry extensions. ***Annals of Statistics***, 53(5), 1913-1937.

3. Kecheng Wei, **Fei Xue**, Qi Xu, Yubai Yuan, Yuxia Zhang, Guoyou Qin, Agaz Wani, Allison Aiello, Derek Wildman, Monica Uddin, Annie Qu (2025). Time-varying mediation analysis for incomplete data with application to DNA methylation study for PTSD. ***Annals of Applied Statistics***, 19(4), 2618-2643.

4. **Fei Xue**, Rong Ma, Hongzhe Li (2025). Semi-supervised statistical inference for high-dimensional linear regression with blockwise missing. ***Statistica Sinica***, 35, 431-456.

5. Jiuchen Zhang, **Fei Xue**, Qi Xu, Jung-Ah Lee, Annie Qu (2024). Individualized dynamic latent factor model with application in mobile health data. ***Biometrika***, 111(4), 1257-1275.

6. **Fei Xue**, Yanqing Zhang, Wenzhuo Zhou, Haoda Fu, Annie Qu (2022). Multicategory angle-based learning for estimating optimal dynamic treatment regimes with censored data. ***Journal of the American Statistical Association***, 117:539, 1438–1451.

7. **Fei Xue**, Xiwei Tang, Grace Kim, Allison Aiello, Karestan Koenen, Sandro Galea, Derek Wildman, Monica Uddin, Annie Qu (2022). Heterogeneous Mediation Analysis on Epigenomic PTSD and Traumatic Stress in a Predominantly African American Cohort. ***Journal of the American Statistical Association***, 117:540, 1669–1683.

8. **Fei Xue**, Annie Qu (2022). Semi-standard partial covariance variable selection when irrepresentable conditions fail. ***Statistica Sinica***, 32, 1881–1909.

9. **Fei Xue**, Annie Qu (2021). Integrating multi-source block-wise missing data in model selection. ***Journal of the American Statistical Association***, 116:536, 1914–1927.

10. Xiwei Tang, **Fei Xue**, Annie Qu (2021). Individualized multi-directional variable selection. ***Journal of the American Statistical Association***, 116:535, 1280–1296.

11. Agaz Wani, Allison Aiello, Grace S Kim, **Fei Xue**, Andrew Ratanatharathorn, Annie Qu, Karestan Koenen, Sandro Galea, Derek Wildman, Monica Uddin (2021). The impact of psychopathology, social adversity and stress-relevant DNAm on prospective risk for post-traumatic stress: A machine learning approach. ***Journal of Affective Disorders***, 282: 894–905.

12. Simona Buetti, **Fei Xue**, Qiawen Liu, Juyoen Hur, Gavin Jun Peng Ng, Wendy Heller (2020). Perceived control in the lab and in daily life impact emotion-induced temporal distortions. ***Timing & Time Perception***, 9(1): 88–122.

13. Grace S Kim, Alicia Smith, **Fei Xue**, Vasiliki Michopoulos, Adriana Lori, Don L. Armstrong, Allison E. Aiello, Karestan C. Koenen, Sandro Galea, Derek E. Wildman, Monica Uddin (2019). Methylomic profiles reveal sex-specific shifts in leukocyte composition associated with post-traumatic stress disorder. ***Brain, Behavior, and Immunity***, 81: 280-–291.

2. In press

1. Yujue Li [G], **Fei Xue**, Bingxuan Li, Yilin Yang, Zirui Fan, Juan Shu, Xiyao Wang, Jinjie Lin, Carlos Copana, Bingxin Zhao. Analyzing bivariate cross-trait genetic architecture in GWAS summary statistics with the BIGA cloud computing platform. ***Journal of the American Statistical Association***, accepted.

2. Antik Chakraborty, **Fei Xue**. Empirical Bayes data integration for multi-response regression. ***Statistica Sinica***, accepted.

3. Submitted (do not include in preparation)

1. Yujue Li [G], **Fei Xue**, Michael G. Levin, Boran Gao, Ali G. Hamedani, Zirui Fan, Jakob Woerner, Yuxin Guo, Cameron Beeche, Juan Shu, Yilin Yang, Xiaochen Yang, Buxin Su, Peristera Paschou, Jinjie Lin, Dokyoon Kim, Walter Witschey, Ruben Gur, Daniel J. Rader, Bingxin Zhao. Network biology of multi-organ imaging genetics uncovers organ-enriched disease mechanisms. Submitted.

2. Huiming Xie [G], **Fei Xue**, Xiao Wang. Nonparametrically Identifiable Deep Latent Variable Models for MNAR Data. Under revision in ***Journal of Machine Learning Research***.

3. Xiaojing Sun [G], Bingxin Zhao, **Fei Xue**. Generalized heterogeneous functional model with applications to large-scale mobile health data. Under revision in ***Journal of the American Statistical Association***.

4. Jin Jin, Bingxuan Li, Xiyao Wang, Xiaochen Yang, Yujue Li [G], Ruofan Wang, Chenglong Ye, Juan Shu, Zirui Fan, **Fei Xue**, Tian Ge, Marylyn D. Ritchie, Bogdan Pasaniuc, Genevieve Wojcik, Bingxin Zhao. PennPRS: a centralized cloud computing platform for efficient polygenic risk score training in precision medicine. Submitted.

5. Rebecca A Deek, **Fei Xue**, James D Lewis, Hongzhe Li. Estimation and false discovery control of multivariate causal effects in omics studies. Submitted.

6. Huiming Xie [G], **Fei Xue**, Xiao Wang. Transition dynamics modeling of pre- trained accelerometry representations for time to diagnosis of Parkinson's disease. Submitted.

7. Yuexia Zhang, Yubai Yuan, **Fei Xue**, Kecheng Wei, Jin Zhou, Annie Qu. Causal mediation analysis of the effect of depression on Alzheimer's disease risk in older adults. Submitted.

4. Non-refereed books and book chapters, etc.

- Huiming Xie [G], **Fei Xue**, Xiao Wang. Generative Models for Missing Data in Applications of Generative AI , Springer, 2024.

C. Invited Lectures

1. March 2026, Colloquium, Indiana University Indianapolis, Indianapolis, IN

   Title: Statistical Methods for Mobile Health Data

2. Feb. 2026, International Seminar on Selective Inference, Online

   Title: High-dimensional statistical inference for linkage disequilibrium score regression and its cross-ancestry extensions

3. Dec. 2025, IMS International Conference on Statistics and Data Science (ICSDS), Seville, Spain

   Title: Federated Transfer Learning for Feature Mismatch

4. Dec. 2025, CMStatistics, London, UK

   Title: Generalized Heterogeneous Functional Model with Applications to Mobile Health Data

5. Oct. 2025, Seminar, Binghamton University, Binghamton, NY

   Title: Statistical Methods for Mobile Health Data

6. July 2025, NSF@75: Advancing Statistical Science for a Data-Driven World, online

   Title: An Empirical Bayes Regression for Multi-tissue eQTL Data Analysis

7. April 2025, Seminar, University of Texas at Dallas, Richardson, TX

   Title: Statistical Methods for Mobile Health Data

8. April 2025, IMSI Research Presentation, Chicago, IL

   Title: Generalized heterogeneous functional model with applications to mobile health data

9. March 2025, ENAR 2025 Spring Meeting, New Orleans, LA

   Title: Time-Varying Mediation Analysis for Incomplete Data with Application to DNA Methylation Study for PTSD

10. March 2025, Seminar, Rice University, Huston, TX

    Title: Statistical Methods for Mobile Health Data

11. Feb. 2025, Seminar, University of Louisville, Louisville, KY

    Title: Statistical Methods for Mobile Health Data

12. Nov. 2024, Undergraduate Women in Science Program (WISP), Purdue University, West Lafayette, IN

    Title: From data to discovery

13. Nov. 2024, SLDS 2024, Newport Beach, CA

    Title: Generalized heterogeneous functional model with applications to mobile health data

14. Sep. 2024, Seminar, University of Kentucky, Lexington, KY

    Title: Statistical Methods for Mobile Health Data

15. Aug. 2024, Conference on Innovative Statistics and Machine Learning for Data Science and AI (IMSL), Bend, OR

    Title: Generalized heterogeneous functional model with applications to mobile health data

16. July 2024, EcoSta 2024 (hybrid), Beijing, China

    Title: High-dimensional statistical inference for linkage disequilibrium score regression and its cross-ancestry extensions

17. June 2024, 2024 WNAR/IMS/Graybill, Fort Collins, CO

    Title: Generalized heterogeneous functional model with applications to mobile health data

18. May 2024, Purdue-Lilly Statistics Seminar, Indianapolis, IN

    Title: High-dimensional statistical inference for linkage disequilibrium score regression and its cross-ancestry extensions

19. May 2024, STATGEN 2024: Conference on Statistics in Genomics and Genetics, Pittsburgh, PA

    Title: High-dimensional statistical inference for linkage disequilibrium score regression and its cross-ancestry extensions

20. June 2023, ICSA Applied Statistics Symposium, Ann Arbor, MI.

    Title: High-dimensional statistical inference for linkage disequilibrium score regression and its cross-ancestry extensions.

21. June 2023, International Purdue Statistics Symposium (IPSS), West Lafayette, IN.

    Title: Individualized dynamic latent factor model with application in mobile health data.

22. May 2023, Midwest Machine Learning Symposium (MMLS), Chicago, IL.

    Title: Deep generative modeling for nonignorable missing data with nonparametric identifiability.

23. May 2023, BFF8, Cincinnati, OH.

    Title: Statistical inference for high-dimensional block-wise missing data.

24. Dec. 2022, CMStatistics (Hybrid), King's College London, UK.

    Title: An Empirical Bayes Regression for Multi-tissue eQTL Data Analysis.

25. Nov. 2022, Lecture Series-Trends in Statistics, National University of Singapore, Singapore.

    Title: Statistical inference for high-dimensional block-wise missing data.

26. Apr. 2022, CCBB Seminar, Indiana University School of Medicine, Indianapolis, IN.

    Title: Statistical inference for high-dimensional block-wise missing data.

27. Apr. 2022, New Challenges and Novel Solutions in Statistics and Data Science, Irvine, CA.

    Title: An Empirical Bayes Regression for Multi-tissue eQTL Data Analysis.

28. March 2022, ENAR Spring Meeting, Houston, TX.

    Title: Heterogeneous Mediation Analysis on Epigenomic PTSD and Traumatic Stress in a Predominantly African American Cohort

29. March 2022, Seminar, Rutgers University, New Brunswick, NJ.

    Title: Statistical inference for high-dimensional block-wise missing data

30. Feb. 2022, Biostatistics Seminar, University of Louisville, Louisville, KY.

    Title: Statistical inference for high-dimensional block-wise missing data

31. Feb. 2022, Statistics Seminar, Indiana University-Purdue University Indianapolis, Indianapolis, IN.

    Title: Heterogeneous Mediation Analysis on Epigenomic PTSD and Traumatic Stress in a Predominantly African American Cohort

32. Jan. 2022, Bioinformatics Seminar, Purdue University, West Lafayette, IN.

    Title: Heterogeneous Mediation Analysis on Epigenomic PTSD and Traumatic Stress in a Predominantly African American Cohort

33. Nov. 2021, EAC-ISBA (Hybrid), Atlantic City, New Jersey, USA and Kunming, Yunnan, China.

    Title: An empirical Bayes regression for multi-tissue eQTL data analysis

34. Oct. 2021, NESS (Hybrid), Kingston, RI, USA.

    Title: Statistical inference for high-dimensional linear regression with blockwise missing data

35. Sep. 2021, The ICSA Applied Statistics Symposium, Online.

    Title: Statistical inference for high-dimensional linear regression with blockwise missing data

36. Aug. 2021, JSM, Online.

    Title: Integrating multi-source block-wise missing data in model selection

37. March 2021, ENAR Spring Meeting, Online.

    Title: Heterogeneous mediation analysis for causal inference

D. Other Presented Papers

- Contributed talks

  1. Aug. 2024, JSM, Portland, OR
     Title: Individualized dynamic latent factor model with application in mobile health data
  2. Aug. 2024, New Researchers Conference (NRC), Corvallis, OR
     Title: Generalized heterogeneous functional model with applications to mobile health data
  3. Aug. 2022, JSM, Washington, DC, USA
     Title: An empirical Bayes regression for multi-tissue eQTL analysis

E. Other Professional Activities

1. 2021–2026, Member of SLDS Student Paper Award Committee, American Statistical Association, online.

2. 2024–2025, Member of the 9th Bayesian, Fiducial, and Frequentist (BFF9) Conference Committee, Purdue University, West Lafayette, IN, USA.

3. December 2024, External Reviewer of Natural Sciences and Engineering Research Council of Canada (NSERC)

4. August 2024, Session Organizer of Novel Statistical Methods for Mobile and Wearable Device Data, JSM, Portland, OR, USA

5. August 2024, Session Chair of New Statistical Methods for High-dimensional Inference, JSM, Portland, OR, USA

6. August 2024, Chair of Funding Panel, New Researchers Conference (NRC), Corvallis, OR

7. March 2024, NSF Panelist of Statistics Program, National Science Foundation

8. June 2023, Session Organizer of Purdue Statistics Alumni for International Purdue Statistics Symposium (IPSS), Purdue University, West Lafayette, IN, USA.

9. June 2023, Session Organizer of Recent Developments in Data Integration for International Purdue Statistics Symposium (IPSS), Purdue University, West Lafayette, IN, USA.

10. March 2023, Panel Organizer of Networking and Career Development, PSAA March Event, online.

11. December 2022, Session Organizer of Recent Developments in Statistical Machine Learning, CMStatistics, London, UK.

12. August 2022, Session Organizer of Recent Developments in Causal Inference, JSM, Washington, DC, USA.

13. March 2022, Panel Organizer of Collaboration and Networking, PSAA March Event, online

F. Interdisciplinary Activities/Collaborations

G. Patents

H. Funding

1. Discussion of support (optional)

2. Award information

The following awards are all external.

- Federal
  - Current
    1. — Funding agent: Natural Science Foundations (NSF) (external funding)
       — Title of Grant: Integrative approaches with applications in eQTL analysis and randomized trials
       — Dates of Funding (beginning to end): Sept. 1, 2022 – Aug. 31, 2026.
       — Total Amount of Award: $125,000.00
       — All Investigators (PIs, Co-PIs, and Co-Is): Fei Xue (PI)
       — Your Role: Principal Investigator
    2. — Funding agent: National Institutes of Health (NIH) (external funding)
       — Title of Grant: Development of Advanced Analytical Tools Integrating External Datasets with MoTrPAC Data: Utilizing Statistical, Machine Learning, and AI Approaches to Uncover Molecular Mechanisms of Physical Activity
       — Dates of Funding (beginning to end): Sept. 22, 2025 – Aug. 31, 2026.
       — Total Amount of Award: $71,168
       — All Investigators (PIs, Co-PIs, and Co-Is): Timothy Reese (PI), Boran Gao (Co-I), Fei Xue (Co-I), Geyu Zhou (Co-I), Yu Michael Zhu (Co-I)
       — Your Role: Co-Investigator
       — If Co-PI, for how much of the total funding are you directly responsible: $11,861.33
  - Past
    1. — Funding agent: Institute for Mathematical and Statistical Innovation (IMSI) (external funding)
       — Title of Grant: Uncertainty Quantification and AI for Complex Systems analysis and randomized trials
       — Dates of Funding (beginning to end): March 3, 2025–May 10, 2025.
       — Total Amount of Award: $13,776.47
       — All Investigators (PIs, Co-PIs, and Co-Is): Fei Xue (PI)
       — Your Role: Principal Investigator
- Industry
  - Current
    1. — Funding agent: Society Of Actuaries (SOA) (external funding)

— Title of Grant: Leveraging AI to Unlock the Potential of Emerging Data for Accelerated Underwriting: A General Framework and Case Study in Wearable-Based Mortality Analytics
— Dates of Funding (beginning to end): Dec. 1, 2025–July 31, 2026.
— Total Amount of Award: $40,000.00
— All Investigators (PIs, Co-PIs, and Co-Is): Jianxi Su (PI), Xiao Wang (Co-I), Fei Xue (Co-I)
— Your Role: Co-Investigator
— If Co-PI, for how much of the total funding are you directly responsible: $13,200

2. — Funding agent: Beck's Superior Hybrids, INC. (external funding)
— Title of Grant: Machine Learning for Predictive Breeding Technology
— Dates of Funding (beginning to end): Aug. 1, 2023 – July 31, 2026.
— Total Amount of Award: $1,921,632.00
— All Investigators (PIs, Co-PIs, and Co-Is): Mark Daniel Ward (PI), Mitch Tuinstra (Co-PI), Luiz Brito (Co-PI), Fei Xue (Co-PI), Raymond Yeh (Co-PI), Ruqi Zhang (Co-PI)
— Your Role: Co-Principal Investigator
— If Co-PI, for how much of the total funding are you directly responsible: $320,143.89

- Past

I. Evidence of Involvement of Students and Postdoctoral Scholars in Research Programs

1. M.S. and Ph.D. students graduated

Name	Graduation date	Degree	Position
Huiming Xie	11/17/2025	PhD	Eli Lilly and Company
Xiaochen Yang	11/28/2025	PhD	Google
Lucy Jewel Kuhl	05/01/2025	MS	–
Zhenyi Wu	08/15/2023	MS	Bioinformatician in Icahn School of Medicine

2. Current graduate students

Co-advised students are indicated with an *.

Name	Start date	Expected completion date	Degree
Ziqi Dong	01/09/2026	08/31/2029	PhD
Huei-Yuan Yao	12/10/2025	08/31/2029	PhD
Mateo Matijasevick*	02/03/2026	08/31/2027	PhD
Hyunwoo Chung	01/08/2024	08/31/2027	PhD
Huayu Tang*	04/15/2023	08/31/2027	PhD
Xiaojing Sun*	08/15/2022	08/31/2027	PhD
Yujue Li*	08/15/2022	08/31/2027	PhD

3. Current and previous undergraduate students with dates and major.

4. Current and previous postdoctoral associates

5. Service on MS/PhD committees with dates

Name	Degree	Start date	Participation	End date
Yunye Liang	MS	03/04/2026	Advisory Committee Member	
Hyunwoo Chung	PhD	01/08/2024	Advisory Committee Chair	
Chaoyu Ding	PhD	12/08/2025	Advisory Committee Member	
Yining Ding	PhD	12/07/2022	Advisory Committee Member	
Ziqi Dong	PhD	01/13/2026	Advisory Committee Chair	
Madison Elizabeth Dunn	MS	10/24/2022	Advisory Committee Member	05/09/2024
Lucy Jewel Kuhl	MS	12/07/2023	Advisory Committee Chair	05/01/2025
Yujue Li	PhD	08/15/2022	Advisory Committee Co-Chair	
Yujue Li	MS	08/15/2022	Advisory Committee Chair	
Zhengming Li	MS	03/23/2022	Advisory Committee Member	
Yu Lin	MS	12/21/2023	Advisory Committee Member	05/05/2025
Chenchen Lu	MS	04/09/2024	Advisory Committee Member	
Mateo Matijasevick	PhD	02/03/2026	Advisory Committee Co-Chair	
Kelly Marie Reagin	MS	12/04/2024	Advisory Committee Member	
Halin Shin	PhD	03/03/2022	Advisory Committee Member	
Xiaojing Sun	PhD	08/15/2022	Advisory Committee Co-Chair	
Shishang Wu	PhD	11/19/2021	Advisory Committee Member	
Huei-Yuan Yao	PhD	12/10/2025	Advisory Committee Chair	
Yizhou Zhang	PhD	12/18/2025	Advisory Committee Member	
Huiming Xie	PhD	07/26/2022	Advisory Committee Co-Chair	11/17/2025
Pengcheng Yang	PhD	11/10/2022	Advisory Committee Member	01/30/2025
Xiaochen Yang	PhD	04/19/2022	Advisory Committee Co-Chair	11/28/2025
Haoyun Yin	PhD	11/01/2021	Advisory Committee Member	10/24/2025
Chen Zhang	MS	01/13/2023	Advisory Committee Member	12/04/2024

IV. Learning

Dr. Fei Xue views teaching statistics as helping students develop a rigorous and lasting way of thinking about data, uncertainty, and real-world decision-making. Rather than emphasizing memorization of formulas, she encourages students to understand the intuition underlying probability, statistical inference, and regression, and to see how these methods can be used to address important questions in areas such as healthcare, economics, and public policy. Her teaching emphasizes the conceptual foundations of statistical methods, the connections among topics across the curriculum, and the practical interpretation of quantitative results.

Equally important to Dr. Xue is creating a classroom environment in which students feel comfortable asking questions, testing ideas, and learning from mistakes. She strives to foster a welcoming and respectful atmosphere that values curiosity, persistence, collaboration, and constructive feedback. Through this approach, she helps students see statistics not as a disconnected collection of techniques, but as a coherent intellectual framework and a skill set that will remain valuable well beyond the classroom.

Since joining Purdue, Dr. Xue has established a strong record of teaching across both graduate and undergraduate levels. Her teaching assignments include STAT 517, Statistical Inference, and STAT 528, Introduction to Mathematical Statistics, both graduate courses that contribute to students' foundational training in theoretical statistics. She has also taught STAT 417, Statistical Theory, multiple times. As a service course for undergraduate students in statistics, mathematics, computer science, engineering, and related fields, STAT 417 plays an important role in the undergraduate curriculum by introducing students to the mathematical basis of statistical inference. Through these teaching assignments, Dr. Xue has contributed to the department's instructional mission in courses that are central to students' development in both theory and application.

A particular strength of Dr. Xue's teaching is her ability to present mathematically rigorous material in a clear and accessible way without sacrificing depth. In courses such as STAT 417, she teaches core topics including inference for parametric families, properties of estimators, Bayesian and maximum likelihood methods, sufficient statistics, hypothesis testing, and the distribution theory underlying commonly used procedures. Because these topics can be abstract and challenging, she emphasizes the relationships among key ideas so that students understand the logic of statistical inference as an integrated whole. This approach helps students move beyond procedural mastery toward deeper analytical understanding.

Dr. Xue's record of student evaluations reflects a consistent pattern of effective teaching. Across multiple semesters and across both graduate and undergraduate courses, her evaluations have been strong in the areas of class preparation, fairness and consistency in evaluation, creation of a welcoming and inclusive classroom environment, and openness to student questions. In several offerings, including STAT 528 in Spring 2023 and multiple sections of STAT 417 in Fall 2024 and Fall 2025, the median evaluation scores reached 5.0 out of 5.0 on most or all reported measures. These results are consistent with her teaching philosophy and indicate that students regard her as well prepared, supportive, fair, and responsive. The consistency of these evaluations across different courses and instructional settings further underscores the quality and reliability of her teaching.

Her sustained teaching of STAT 417 is especially notable because it demonstrates continued contribution to a key undergraduate course that serves a broad student population. At the same time, her teaching of STAT 517 and STAT 528 shows her engagement with graduate education and with the department's responsibility to provide students with a strong foundation in statistical theory. Together, these assignments reflect both breadth and depth in her instructional contributions.

In addition to her formal classroom teaching, Dr. Xue has contributed to undergraduate education through outreach activities that encourage student engagement with statistics and data science.

For example, she gave the talk "From Data to Discovery" for Purdue's Undergraduate Women in Science Program, helping promote student success and interest in data-driven inquiry. This effort complements her classroom teaching by extending her educational impact beyond enrolled students and supporting broader participation in quantitative fields.

Overall, Dr. Xue has developed a thoughtful and effective teaching profile characterized by rigor, clarity, inclusiveness, and sustained commitment to student learning. She teaches foundational statistical material in a way that promotes both technical competence and conceptual understanding, and she does so in a classroom environment that supports student engagement and growth. Her teaching record demonstrates meaningful contributions to both undergraduate and graduate education and aligns well with the instructional mission of the Department of Statistics and the broader expectations for promotion.

A. Teaching Assignments at Purdue

Online courses are indicated with an *.

Semester and Year	Course Number, Credit Hour, Type	Title of Course	Number of Students	Student Classification
S 2026	STAT 417, 3 cr, lecture	Statistical Theory	49	Soph & Jr & Sr
F 2025	STAT 417, 3 cr, lecture	Statistical Theory	48	Soph & Jr & Sr
F 2025	STAT 417, 3 cr, lecture	Statistical Theory	49	Soph & Jr & Sr
F 2024	STAT 417, 3 cr, lecture	Statistical Theory	50	Soph & Jr & Sr
F 2024	STAT 417, 3 cr, lecture	Statistical Theory	50	Soph & Jr & Sr
S 2024	STAT 417, 3 cr, lecture	Statistical Theory	48	Soph & Jr & Sr
S 2024	STAT 417, 3 cr, lecture	Statistical Theory	45	Soph & Jr & Sr
S 2023	STAT 528, 3 cr, lecture	Introduction to Mathematical Statistics	21	Gr
F 2022	STAT 517, 3 cr, lecture	Statistical Inference	28	Gr
S 2022	STAT 517 - WNG, 3 cr, lecture	Statistical Inference	19	Gr
S 2022	STAT 517* - EPE, 3 cr, lecture	Statistical Inference	13	Gr
F 2021	STAT 517, 3 cr, lecture	Statistical Inference	12	Gr

B. Selected Discussion of Courses

STAT 417 - Statistical Theory: This course is a service course for undergraduate students in mathematics, statistics, computer science, engineering, and some other fields. The course provides an introduction to the mathematical theory of statistical inference, emphasizing inference for standard parametric families of distributions, properties of estimators, Bayes and maximum likelihood estimation, sufficient statistics, properties of test of hypotheses, and distribution theory for common statistics based on normal distributions.

C. Course Evaluations

1. Student Evaluation

We provide the median of responses to the following four questions:

- My instructor seems well-prepared for class.
- The instructor is fair in evaluating my performance in the course.
- The instructor created an inclusive classroom environment.
- The instructor effectively answers students' questions.

Online courses are indicated with an *.

Semester and Year	Course	Well-Prepared	Fair and Consistent	Welcoming and Inclusive	Open to Questions
S 2026	STAT 417	5.0/5.0	5.0/5.0	4.5/5.0	5.0/5.0
F 2025	STAT 417	5.0/5.0	5.0/5.0	4.5/5.0	5.0/5.0
F 2025	STAT 417	5.0/5.0	5.0/5.0	4.5/5.0	4.0/5.0
F 2024	STAT 417	5.0/5.0	5.0/5.0	5.0/5.0	5.0/5.0
F 2024	STAT 417	4.0/5.0	4.0/5.0	4.0/5.0	4.0/5.0
S 2024	STAT 417	4.0/5.0	4.0/5.0	4.0/5.0	4.0/5.0
S 2024	STAT 417	4.0/5.0	5.0/5.0	4.0/5.0	4.0/5.0
S 2023	STAT 528	5.0/5.0	5.0/5.0	5.0/5.0	5.0/5.0
F 2022	STAT 517	4.0/5.0	5.0/5.0	4.0/5.0	4.0/5.0
S 2022	STAT 517 - WNG	5.0/5.0	5.0/5.0	5.0/5.0	5.0/5.0
S 2022	STAT 517* - EPE	4.0/5.0	4.0/5.0	4.0/5.0	4.0/5.0
F 2021	STAT 517	5.0/5.0	5.0/5.0	5.0/5.0	5.0/5.0

2. Peer Evaluation

D. Other Contributions to Undergraduate Education

- Gave a talk on "From data to discovery" for Purdue Undergraduate Women in Science Program (WISP) to promote success of undergraduate students.

V. Engagement and Service

A. Discussion of Engagement

Dr. Fei Xue's engagement activities connect her scholarly work in statistics and machine learning with needs in industry, interdisciplinary science, and student outreach, thereby advancing Purdue's land-grant mission. Her work emphasizes the development of rigorous analytical methods for real-world problems and the translation of statistical research to settings outside the university.

An important example is her collaboration with Beck's Superior Hybrids company on machine learning for predictive breeding technology. This partnership applies modern statistical and machine learning methods to problems of practical importance in agriculture and demonstrates how methodological research can contribute to industry innovation. Her mentoring of a research assistant in this collaboration further extends the impact of this work by involving students in applied, externally engaged research.

Dr. Xue also collaborates with researchers at institutions including the University of Pennsylvania, the University of Pittsburgh, Rutgers University, Harvard University, and the University of California, Irvine. Through these partnerships, her methodological expertise contributes to interdisciplinary research in biomedical and health-related areas that require new tools for complex and high-dimensional data. These collaborations broaden the reach of Purdue's research and strengthen connections between statistical methodology and important scientific questions.

In addition, Dr. Xue contributes to Purdue's educational outreach mission through activities such as her talk, "From Data to Discovery," for Purdue's Undergraduate Women in Science Program. This effort helped promote student interest in statistics and data-driven research and supported the success of undergraduate students.

Overall, Dr. Xue's engagement demonstrates how her scholarship serves needs both inside and outside the academy. Through industry collaboration, interdisciplinary research partnerships, student mentoring, and outreach, she extends the impact of her work beyond the classroom and the university while supporting Purdue's commitments to discovery, education, and service.

a. Engagement with Partners

- Collaborating with Beck's Superior Hybrids company on machine learning for predictive breeding technology
- Mentoring a research assistant for Beck's Superior Hybrids company
- Collaborating with University of Pennsylvania, University of Pittsburgh, Rutgers University, Harvard University, University of California, Irvine, and some other universities on statistical research

b. Individuals Mentored Through Engagement Activities

c. Impact of the Scholarship of Engagement

d. Technology Transfer or Commercialization Results of Engagement

e. Other Engagement Activities

B. Discussion of Service

Dr. Fei Xue has made sustained service contributions at the departmental and professional levels. Within the Department of Statistics, she currently serves as Chair of the Colloquium Committee, a member of the Graduate Program Committee, a member of the Talent-Based Hires Committee, and Chief Executive Officer of the Purdue Statistics Alumni Association. She has also contributed to departmental initiatives through service on the Data Science Search Committee, the External Review Committee, and the organizing and academic program committees for the International Purdue Statistics Symposium, as well as by organizing alumni events.

Her professional service reflects growing national and international recognition in statistics. She served as an NSF panelist in 2024 and as an external reviewer for the Natural Sciences and Engineering Research Council of Canada in 2024. She has provided extensive manuscript review service for leading journals in statistics, biostatistics, and machine learning, including the *Journal of the American Statistical Association*, *Journal of the Royal Statistical Society: Series B*, *Annals of Statistics*, *Annals of Applied Statistics*, *Journal of Machine Learning Research*, *Biometrics*, *Bioinformatics*, and *Statistics in Medicine*, among others. Her reviewing activity has been substantial and consistent across recent years.

Dr. Xue has also contributed to the profession through service to scholarly societies and conference communities. She has served on the ASA Statistical Learning and Data Science Student Paper Award Committee from 2021 to 2026, organized invited sessions at JSM and CMStatistics, chaired the Funding Panel at the 2024 New Researchers Conference, and served as a poster judge at ICSA and MMLS. In addition, she has supported the broader academic community through writing recommendation letters for students and for professional award nominations.

Overall, Dr. Xue's service record shows consistent engagement with departmental governance, student and alumni connections, peer review, professional societies, and the broader statistical community.

a. Department

F2025–current	Chair of Colloquium Committee
F2024–current	Member of Graduate Program Committee
F2023–current	Member of Talent-Based Hires Committee
F2021–current	Chief Executive Officer of Purdue Statistics Alumni Association (PSAA)
F2024–S2025	Member of the 9th Bayesian, Fiducial, and Frequentist (BFF9) Conference Committee
F2022– S2023	Member of Local Organizing Committee for International Purdue Statistics Symposium (IPSS)
F2022– S2023	Member of Academic Program Committee for International Purdue Statistics Symposium (IPSS)
F2021 – S2022	Member of Data Science Search Committee
S2022	Member of External Review Committee
S2022	Organizer of Purdue Statistics Alumni Association (PSAA) March Event.

b. College

c. University

d. Professional

- NSF Panelist of Statistics Program in 2024
- External Reviewer of Natural Sciences and Engineering Research Council of Canada (NSERC) in 2024
- Reviewed 10 scientific manuscripts in 2021, 19 scientific manuscripts in 2022, 13 manuscripts in 2023, 9 manuscripts in 2024, 16 manuscripts in 2025 for numerous professional journals including:
    1. Journal of the American Statistical Association
    2. Journal of the Royal Statistical Society: Series B
    3. Annals of Statistics

4. Annals of Applied Statistics
5. Journal of Machine Learning Research
6. Electronic Journal of Statistics
7. Journal of Nonparametric Statistics
8. Journal of Computational and Graphical Statistics
9. PLOS One
10. Biometrics
11. Bioinformatics
12. Biostatistics
13. The Canadian Journal of Statistics
14. Statistics in Medicine
15. Statistica Sinica
16. Lifetime Data Analysis
17. Science China Mathematics
18. Journal of Multivariate Analysis

- Member of SLDS Student Paper Award Committee at the American Statistical Association (ASA) from 2021 to 2026.
- Organized sessions in JSM and CMStatistics in 2022–2024.
- Chaired the Funding Panel in New Researchers Conference (NRC) in 2024.
- Wrote a recommendation letter for a nomination of the Florence Nightingale David Award in 2024, and for a nomination of the Elizabeth L. Scott Award in 2025.
- Wrote recommendation letters for four students in 2022, four students in 2023, five students in 2024, four students in 2025, and four students in 2026.
- Poster Session Judge in ICSA 2020 and MMLS 2023.

e. Consulting Activities

f. Other Service Activities

VI. Mentoring

Dr. Fei Xue views mentoring as an important part of supporting students' academic progress, research development, and professional growth. Her mentoring approach is active, responsive, and individualized, with emphasis on helping students build confidence, develop independence, and navigate key academic and career decisions. She works to create an environment in which students feel comfortable asking questions, discussing challenges, and refining their ideas through regular guidance and constructive feedback.

Her mentoring includes helping students begin research projects, develop the skills needed to read the literature and make research progress, and identify appropriate next steps in their academic development. She has mentored undergraduate students through research advising and career preparation, including support for graduate school applications and recommendation letters. At the graduate level, she has provided guidance through both research advising and faculty mentoring, supporting students in research planning, academic adjustment, and professional development. She has also met with prospective graduate students and contributed to their introduction to the department and its research environment.

Across these activities, Dr. Xue emphasizes clarity, encouragement, and steady intellectual growth. Her mentoring extends beyond technical instruction to include broader support for student success and professional advancement. Overall, her record reflects a sustained commitment to thoughtful and effective mentorship of students at multiple stages of development.

A. Undergraduate students

- Advised two undergraduate students (Siqi Chen and Qinkai Yu) on research projects, and wrote recommendation letters for them. Helped them start and make good progress in their research projects. Also helped them apply for PhD programs. Siqi got a PhD position at University of Pennsylvania. Qinkai got a PhD position at University of Exeter.

B. Graduate students

- Advised a master student (Tsu-Hao Fu) in Department of Statistics at Purdue to start and make good progress in his research project
- Served as a faculty mentor for Bochang Yang in 2025, Yizhou Zhang, Ximing Ding in 2024, for Lucy Jewel Johnson in 2023, and for Yujue Li and Xiaojing Sun in 2022
- Met with prospective graduate students in March 2023

C. Postdoctoral Scholars

D. Faculty members